

Spotify Music Data Analysis

SQL-Based Exploratory & Business Intelligence Project

Project Overview

This project performs an end-to-end SQL analysis of a Spotify music dataset that combines streaming platform metrics with YouTube engagement data. The analysis spans exploratory data analysis, business intelligence queries, and advanced analytics using window functions and CTEs.

The dataset captures track-level attributes including audio features (danceability, energy, tempo, valence), streaming numbers, YouTube views/likes/comments, licensing status, and platform play preferences — making it ideal for cross-platform performance analysis.

Tech Stack

Component	Details
Database	PostgreSQL
Language	SQL (Structured Query Language)
Techniques	DDL, DML, Aggregations, Subqueries, CTEs, Window Functions
Dataset	Spotify + YouTube cross-platform music metrics
Tool	pgAdmin / Any PostgreSQL client

Dataset Schema

The spotify table contains 24 columns organized across three categories:

Track Identity

- artist — Performing artist name
- track — Song title
- album — Album name
- album_type — Classification (single / album)

Audio Features

- danceability, energy, loudness, speechiness
- acousticness, instrumentalness, liveness
- valence (mood positivity), tempo, duration_min

Engagement Metrics

- views, likes, comments — YouTube metrics
- stream — Total Spotify streams
- licensed, official_video — Content flags
- most_played_on — Platform preference (Spotify / YouTube)

Analysis Breakdown

Phase 1 — Exploratory Data Analysis (EDA)

- Counted total records and distinct artists to understand dataset scale
- Identified unique album types and channel categories
- Detected and removed null/zero-duration tracks for data quality
- Profiled max/min values across views, likes, comments, and streams
- Segmented licensed vs. non-licensed tracks and platform distribution

Phase 2 — Basic Business Problems

#	Business Question	SQL Concept
1	Tracks with 1B+ streams	WHERE filter
2	Albums with their artists	DISTINCT + ORDER BY
3	Total comments for licensed tracks	SUM + WHERE
4	Tracks from 'single' album type	Categorical filter
5	Track count per artist	GROUP BY + COUNT

Phase 3 — Medium Level Business Problems

#	Business Question	SQL Concept
6	Avg danceability per album	AVG + GROUP BY + ROUND
7	Top 5 highest energy tracks	ORDER BY + LIMIT
8	Views & likes for official videos	SUM + WHERE boolean
9	Total views per album's tracks	GROUP BY multi-column
10	Tracks streamed more on Spotify vs YouTube	CASE WHEN + COALESCE + subquery

Phase 4 — Advanced Analytics

#	Business Question	SQL Concept
11	Top 3 most-viewed tracks per artist	DENSE_RANK() window function + CTE
12	Tracks with above-average liveness	Correlated subquery + AVG
13	Energy range (max–min) per album	CTE + derived column arithmetic

Key Insights & Findings

- Cross-platform comparison revealed which tracks dominate on Spotify vs YouTube, enabling platform-specific marketing strategies.
- Window functions with DENSE_RANK identified top-performing tracks per artist without data duplication.
- Audio feature analysis (danceability, energy, liveness) demonstrated measurable correlation with streaming performance.
- Licensed content showed significantly higher engagement metrics in comment and view volumes.
- Energy variance analysis per album helped identify musically diverse vs. consistent albums.

SQL Skills Demonstrated

Skill Category	Techniques Used
Data Definition (DDL)	CREATE TABLE, DROP TABLE, data types
Data Manipulation (DML)	INSERT, DELETE, SELECT
Aggregations	COUNT, SUM, AVG, MAX, MIN, ROUND
Filtering & Sorting	WHERE, ORDER BY, LIMIT, DISTINCT
Grouping	GROUP BY with multi-column aggregations
Subqueries	Scalar subqueries for dynamic comparisons
Conditional Logic	CASE WHEN, COALESCE for null handling
CTEs	WITH clause for readable multi-step logic

Window Functions	DENSE_RANK() with PARTITION BY + ORDER BY
Data Cleaning	Detecting and removing invalid records